

Introduction to Microeconomics

Lecture 8—Consumer Theory: The Budget Set

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The budget set and the budget line for the consumer.

- Goal: represent what is feasible to consume given market prices and income.
- Consumption space: $X = \mathbb{R}_+^n$, n goods, indexed x_i , $i = 1, \dots, n$.
- Prices $p_1, \dots, p_n$ for each good, all non-negative.
- Consumer takes prices as given, i.e., consumer cannot “bargain” for a different price.
- The budget set is the set of all **feasible** consumption bundles given prices and income I of the consumer:

$$B(p_1, \dots, p_n, I) = \{(x_1, \dots, x_n) : p_1x_1 + p_2x_1 + \dots p_nx_n \leq I\}$$

- Intuitively: Total expenditures must be less than or equal to Income.
- When $p_1x_1 + \dots + p_nx_n = I$, this corresponds to the **budget hyperplane**, and is the set of consumption bundles that are feasible and exhaust the consumer's income.

The two-good case

- Suppose we are only considering two goods, i.e., $X = \mathbb{R}_+^2$, goods x_1 and x_2 .
- Prices are p_1 and p_2 and income I .
- The budget set is described by the simple inequality: $p_1x_1 + p_2x_2 \leq I$.
- The budget line is $p_1x_1 + p_2x_2 = I$ is the set of feasible consumption bundles which exhaust the consumer's Income.

The slope of the budget line

Non-linear budget constraints